

VIVEK SHUKLA

Backend Developer | Python & Django Developer | Cloud & DevOps Engineer

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PROFESSIONAL SUMMARY

Results-driven Computer Science fresher with hands-on experience in Python, Django, REST API development, Node.js, Express.js, MongoDB, SQL, AWS (EC2, S3), Linux administration, CI/CD pipelines, Shell Scripting, and DevOps practices. Skilled in designing and deploying scalable backend systems, implementing CRUD operations, integrating third-party APIs, and building modular, production-ready backend architectures. Proficient in cloud deployment via AWS and Vercel, with practical experience in Git, GitHub, Postman API testing, and agile development workflows. Recognised by NASA as a 'Galactic Problem Solver' for delivering a high-availability, cloud-hosted, REST API-integrated web application. Strong problem-solving, analytical thinking, and communication skills with a passion for building efficient, maintainable backend infrastructure. Actively seeking fresher or internship-level opportunities as a Backend Developer, Python Developer, Django Developer, Cloud Engineer, DevOps Engineer, or Software Engineer.

WORK EXPERIENCE

Cloud Computing Intern

Acmegrade Pvt Ltd | Dec 2024 – Jan 2025

- Engineered three production-grade conversational AI bots using AWS Lex, implementing backend logic for intent recognition, slot filling, multi-turn dialogue management, and automated customer-service workflows — improving response efficiency and user engagement by streamlining query resolution.
- Developed a 'Car Buying Assistant' backend system integrating API-driven business logic to dynamically query and filter a catalogue of 20+ car models based on price, colour, and delivery preferences; applied CRUD operations and JSON-based data exchange to deliver structured API responses.
- Designed backend processing logic and conversation flows for an Online Shopping bot and a Railway Ticket Booking system (replicating IRCTC General/Tatkal quota logic), incorporating authentication flows, middleware-based slot validation, async/await request handling, and scalable intent routing for concurrent user sessions.
- Conducted rigorous functional and integration testing using Postman to validate REST API endpoints and intent accuracy across edge cases; optimised backend response pipelines, reducing error rates and improving reliability in production-simulated cloud environments.

PROJECTS

Spacifyer – NASA Space Apps Challenge

spacifyer-a-geomagnetic-experience.vercel.app | Oct 2024

- Developed a full-stack, cloud-hosted web application that consumes NASA's space-weather REST APIs via HTTP GET requests, parsing real-time JSON responses to visualise geomagnetic data — demonstrating end-to-end API integration, backend data-processing pipelines, and dynamic frontend rendering.
- Implemented a modular backend architecture with clean separation of concerns (API layer, data transformation, presentation), enabling maintainability, scalability, and adherence to production-ready development standards.
- Deployed the application on Vercel with automated CI/CD pipelines configured for continuous deployment from GitHub, ensuring zero-downtime releases, high availability, and global CDN-backed accessibility.
- Awarded 'Galactic Problem Solver' by NASA International Space Apps Challenge 2024 — globally recognised for impactful engineering solutions addressing Earth and Space science challenges.

Titanic Survival Prediction – Machine Learning Classification System

Aug 2025 – Oct 2025

- Built an end-to-end machine learning pipeline in Python (Pandas, Scikit-learn) performing full ETL operations: handling missing values, feature engineering (Title extraction, FamilySize aggregation), data normalisation, and train/test splitting to prepare a production-ready dataset.
- Implemented and benchmarked Logistic Regression, Random Forest, and XGBoost classifiers for binary classification; applied Randomised SearchCV hyperparameter tuning to optimise model accuracy and generalisation performance.
- Structured the codebase using modular Python scripts, clear inline documentation, and full version control via Git and GitHub, following software engineering best practices for collaborative, maintainable development.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Shell Scripting (Bash)

Backend Technologies: Django, REST API Design and Development, Node.js, Express.js, CRUD Operations, Middleware, Authentication, JSON, Async/Await, API Integration, Backend Architecture

Databases: MongoDB, SQL (Basics), Database Integration, Schema Design

Cloud and DevOps: AWS EC2, AWS S3, AWS Lex, Cloud Computing, Linux Administration, CI/CD Pipelines, Vercel Deployment, DevOps Fundamentals, Shell Scripting, Automation

API and Development Tools: Git, GitHub, Postman (API Testing), VS Code

Frontend / Web: HTML5, CSS3, JavaScript

AI / Machine Learning: Scikit-learn, Pandas, XGBoost, Random Forest, Logistic Regression, Predictive Modelling, Feature Engineering

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

School of Computer Science and Engineering, Sandip University, Nashik | Sep 2022 – Sep 2026 (Expected)

- Relevant Coursework: Data Structures and Algorithms, Cloud Computing, Operating Systems, Distributed Systems, Database Management Systems, Object-Oriented Programming.
- Thesis: Innovations in Sustainable Engineering Practices.

CERTIFICATIONS

- Artificial Intelligence and Machine Learning – Learn Flu (AICTE Approved)
- Entrepreneurship Essentials – NPTEL, IIT Kharagpur | Score: 50%

AWARDS AND ACHIEVEMENTS

- Galactic Problem Solver – NASA International Space Apps Challenge 2024: Globally recognised for engineering a cloud-hosted, REST API-integrated geomagnetic data visualisation platform with scalable, production-ready backend architecture and automated CI/CD deployment.
- Public Speaking – Active speaker on Cloud Computing and Cyber Security trends at college-level technical events, workshops, and seminars.

SOFT SKILLS

Problem Solving | Analytical Thinking | Team Collaboration | Communication | Adaptability | Quick Learning | Time Management | Public Speaking

LANGUAGES KNOWN

English – Professional Proficiency | **Hindi** – Native / Full Professional Proficiency